

A co-evolutionary perspective on humans and *Mycobacterium tuberculosis* in the era of systems biology

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This Review Article explores the intricate relationship between humans and *Mycobacterium tuberculosis* (Mtb), providing an additional perspective on TB disease. Specifically, this review focuses on the utilization of systems-level approaches to study TB, while highlighting challenges in the frameworks used to identify the relevant immunologic signals that may explain the clinical spectrum of disease. The work could be further enhanced by better defining key terms that anchor the review, such as "unified mechanism" and "immunological route." This review will be of interest to immunologists as well as those interested in evolution and host-pathogen interactions.

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Abstract

Tuberculosis is once again the most fatal global infectious disease and has killed many more humans than any other pathogen. Despite the identification of *Mycobacterium tuberculosis* (Mtb) over 140 years ago, we have yet to control the epidemic. A central issue is the complexity of the host-pathogen interaction, with diverse immunological routes leading to tuberculosis disease. This intricate relationship results from the prolonged co-evolution of the pathogen with humans, potentially due to Mtb exposure giving a survival advantage. Emerging methodologies, such as single cell and spatial transcriptomics, offer a golden opportunity to understand the immunology unpinning this host-pathogen interaction at unprecedented resolution. However, these analyses will be fundamentally flawed if they do not consider the intricacies of human Mtb infection. Here, we propose that attempts to find single unifying immunological mechanisms of tuberculosis are hindering progress, and we

must embrace the complexity of multiple paths to disease to allow the systems biology era to deliver transformative solutions.

Introduction

The study of tuberculosis (TB) pioneered infectious disease research in the modern scientific era, leading to the formulation of Koch's postulates demonstrating that an illness can have an infectious origin (1). *Mycobacterium tuberculosis* (Mtb) infects and survives within macrophages, subverting the host immune response by multiple mechanisms including inhibition of phagosome maturation and downregulation of antigen presenting molecules, leading to the formation of complex immune aggregates, known as granulomas (2). Even though TB was the first definitively identified infection, it remains the world's deadliest infectious disease despite well over 100 years of research (3). This contrast raises a critical question: why is Mtb proving so resistant to human efforts to control it?

Mtb has eluded attempts to develop a fully protective vaccine, despite a partially effective vaccine being available since the 1920's. *Mycobacterium bovis* BCG was developed by sequential culture of *M bovis* and protects children against disseminated TB but has limited protection against adult disease (4, 5). Although exciting progress has been made with a vaccine that reduces TB disease by approximately 50% in a phase II study (6), currently the immune mechanisms underpinning protection have not been identified. Subsequently, a major trial of an alternative vaccine showed no efficacy in preventing recurrence. Indeed, despite being highly immunogenic, the relapse rate tended to be higher in the vaccine group (7). Similarly, repeat BCG vaccination does not increase protection, despite inducing a strong Mtb-specific CD4 T cell response (8).

These difficulties highlight the urgent need to understand the host-pathogen interaction more fully. We have insufficient knowledge of key steps in disease progression to develop transformative interventions. Currently, almost all investigations into the immune response to Mtb are based on the premise that a unifying disease mechanism exists, and this can be understood through reductionist scientific approaches. However, clinical observations demonstrate that there are multiple paths to TB, and so seeking to define a single mechanism is likely to be flawed.

Immunological insights from historic and recent clinical observations

TB has co-evolved with humans for millennia, with some estimates suggesting up to 70,000 years (9). The field of paleoarchaeology, for example, provides extensive evidence of TB from the early neolithic period in the Middle East, with approximately 5% of skeletons from a 10,000-year-old village showing signs of TB (10). This was just before animal domestication and pottery, in hunter-gatherers who built stone houses, and so Mtb was already successfully transmitting in humans before the subsequent population growth that occurred with farming (10). Potentially, to survive in relatively small hunter-gatherer communities Mtb may have needed to have reduced virulence to avoid excessive deaths and a latent period to permit sustainable transmission in low population numbers (11).

Mtb successfully persisted over the ages and then flourished in the more dense populations that occurred with the industrial revolution (12), giving rise to the modern TB era of high transmission approximately 250 years ago. The fundamental cause of TB remained unknown until Koch's seminal work (13). Early investigations demonstrated that the first Mtb infection point was the lung base, while Mtb exits from the apices of the lungs (13). This life cycle must involve

several distinct host/pathogen interactions, as initially immune evasion is required for Mtb to survive, but then later immune engagement is necessary to cause the inflammation and lung destruction needed to optimise transmission (14). Notably, most people (approximately 90%) initially infected with Mtb never progress to active, clinical disease (3). In addition, in the pre-antibiotic era, the progression and regression of different lesions in the same individual were observed on chest radiographs, and one third of patients with active TB disease self-healed, showing that the host-pathogen interaction is finely balanced at all stages of infection (12).

Modern immunological techniques and the development of biologic therapies that target specific immune processes have provided extensive insight into TB disease mechanisms. The greatly increased occurrence of TB in the context of HIV co-infection, for example, highlighted immunodeficiency as a major driver of disease (15). Similarly, the occurrence of TB after anti-TNF- α therapy for autoimmune disease confirmed the importance of TNF- α in control of latent infection (16). Furthermore, genetic investigations have identified numerous immunodeficiencies via studies of Mendelian Susceptibility to Mycobacterial Diseases (MSMD), with mutations typically along the IL-12/IFN- γ /STAT signalling pathway (17–19). With less clearly defined immunologic mechanisms, malnutrition is a significant risk factor for TB (20), and food supplementation reduces TB incidence in contacts (21). Therefore, diverse immune deficiencies can lead to active TB.

The vast majority of patients who develop TB, however, have no clear identifiable immunodeficiency. Indeed, Comstock's seminal study from the 1970s showed that children from Haiti with the strongest recall responses to Mtb antigens actually had the greatest subsequent risk of developing TB (22). These observations have been replicated in more recent studies using IFN- γ release assays (IGRAs) in response to TB antigens, where higher IFN- γ production associates with increased risk of progressing to disease in both children and adults (23, 24). TB is commonest in young adults in their immunological prime, and more frequent in males than females, characterised by an excessive inflammatory response (25). The implication that TB can also be caused by immune excess is now supported by recently introduced cancer immunotherapies (26). Anti-PD-1 treatment, which activates the immune response and represents the immunological opposite to anti-TNF therapy, should control TB if immunodeficiency were the critical component. However, anti-PD-1 treatment can lead to rapid reactivation of latent TB infection, first identified in case reports (27). This finding is supported by studies in mice (28, 29), the non-human primate (30) and 3D cellular models (31), and ultimately has been validated by patient registry studies (32, 33). Similarly, type II diabetes is associated with an increased risk of TB (20), characterized by a hyper-inflammatory immune response (34). Therefore, diverse clinical evidence demonstrates that there are multiple immunological routes to TB disease (35) (**Figure 1**).

Even when TB develops in the face of a “normal” immune system, there are likely to be many different subgroups that have not yet been identified due to limitations in standard immunological profiling. A quarter of the world's population are thought to be exposed to Mtb (36), and so the vast majority of those infected with Mtb remain healthy lifelong (3), while those who progress to active TB likely represent distinct outlier populations. The diverse causes of active TB, such as anti-TNF, MSMD, HIV, diabetes and anti-PD1, demonstrate that patients do not reach TB by a single route, but instead can lose the immunological balance that controls Mtb by multiple paths. This may explain why genetic studies have generally failed to find consistent predispositions. Evidence of heritability can be demonstrated but is often hard to validate in different populations (37). Mutations in the macrophage endosomal protein NRAMP1, for example, were shown to associate with TB in an early seminal study (38), but since then few consistent traits of genetic susceptibility to TB have been identified (39). Similarly, studies of individuals who resist Mtb infection despite recurrent exposure generates a diverse list of potentially protective features, including T cell subsets or activation (40–42), TNF- α responses (43), antibody activity (44) and innate immune training (45). These human observations are supported by murine

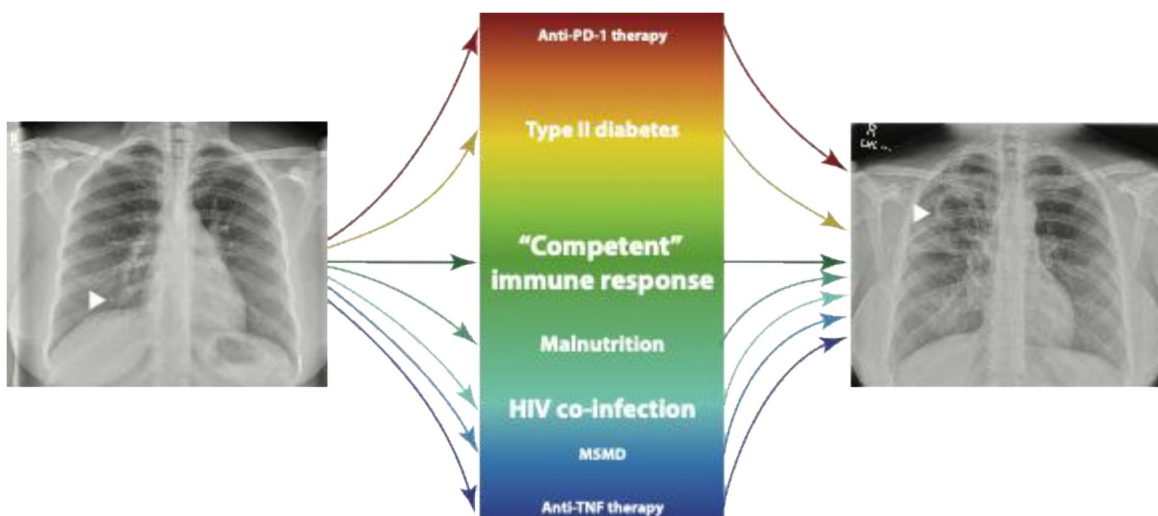


Figure 1

Many routes can lead to TB via opposing immunological extremes.

Generally, these can be viewed as immune deficiencies or immune excess, but the majority of patients who develop TB are relatively young with a competent immune response. Though not quantitative, font size relates to contribution to global incidence. Left arrow: Ghon focus at lung base; Right arrow: cavity at lung apex.

genetic experiments, which show that increased susceptibility to Mtb can result from a wide range of immunological alterations (46 [↗](#)). However, despite this evidence of diversity, the TB field continues to seek a single unifying explanation for disease progression, which is incompatible with clinical observations.

Ultimately, to transmit efficiently, Mtb needs to cause pulmonary disease to then spread by airborne droplets (47 [↗](#)), and for Mtb it does not matter the route taken, as long as it ends at pulmonary TB. Experimentally, TB is a fundamentally challenging disease to model, as the interaction is prolonged and Mtb is an obligate human pathogen (48 [↗](#)). The primary driver of advances in immunology in the last decades has been transgenic mice (49 [↗](#)), but the mouse model of TB does not accurately reflect human disease (50 [↗](#)). Disease heterogeneity has been highlighted in describing clinical TB endotypes observed during active disease (51 [↗](#)), but less attention has been given to the different immunological pathways that can lead to the same disease phenotype.

Mtb's single successful colonisation of humans

Just as new tools are providing insights into the complexity of human TB progression, advances in mycobacterial genomics are highlighting the unique nature of the human-Mtb relationship (52 [↗](#)). Mtb is a near clonal organism, with evidence suggesting that there has been only one successful and sustained penetration into the human population (52 [↗](#), 53 [↗](#)). The entire spectrum of Mtb strains globally only differs by a total of approximately 2,000 SNPs (52 [↗](#)). This genetic conservation has persisted even though Mtb has been flourishing in human populations since the industrial revolution, suggesting that Mtb was already close to being optimised for human hosts. Further evolution may have been to increase transmission within specific populations as humans diverged genetically (53 [↗](#)).

A very similar organism, *M canetti*, can cause disease but cannot transmit from human to human (54 [↗](#)). Similarly, *M bovis* is 99.9% identical to Mtb (55 [↗](#)), but has never achieved sustained human-to-human transmission, despite many millions of human exposures during the pre-pasteurisation era and common lymph node infections (53 [↗](#)). Therefore, human TB is caused exclusively by Mtb, unlike other very closely related mycobacteria, which maintain infection cycles in other higher mammals but not humans (53 [↗](#)). Clues to the key pathogenic mechanisms may lie in the differences between mycobacterial species, but ultimately similarities between Mtb strains must be critical to their ongoing success. For example, the hyper conservation of T cell epitopes may be evidence that Mtb manipulates the host immune response to favour disease and transmission (56 [↗](#)). Given the size of the Mtb genome, identifying the critical conserved features will be challenging, especially as half of its genes still lack a known function 25 years after the genome was first sequenced (57 [↗](#)).

One highly intriguing proposition is that latent TB may itself give humans an evolutionary advantage (57 [↗](#)). Exposure to Mtb modifies innate immune training (58 [↗](#)), and so a mechanism whereby Mtb could protect from other fatal infections is plausible. Similarly, *M bovis* BCG, a live-attenuated strain of *M bovis*, causes innate immune training (59 [↗](#)), suggesting this effect is common to Mtb and BCG. BCG vaccination reduces mortality much more significantly than can be explained by the effect on TB incidence alone (60 [↗](#)), with a protective effect confirmed in diverse studies (61 [↗](#)). For example, BCG reduces viral infections in infants in Africa (62 [↗](#)) and experimentally challenged adults (63 [↗](#)), and associates with improves survival in Europe (64 [↗](#)). Together, these observations strongly imply that mycobacterial infection protects from other infectious causes of death.

Mtb almost certainly first colonised humans in East Africa (53 [↗](#)), potentially about 70,000 years ago (9 [↗](#)). Several human migrations out of Africa are thought to have preceded this date, but all ultimately became extinct, with the first sustained human dispersal 60 – 70,000 years ago (65 [↗](#), 66 [↗](#)). The primary selective pressure in these early communities would have been infectious disease (67 [↗](#)). This raises a novel hypothesis that a survival advantage for the first successful human migrants out of Africa was Mtb circulating in the community, reducing mortality from other infectious diseases and thereby enabling sustainable population growth.

Mtb transmission may have benefitted humans by increasing innate immune resistance to infection at the cost of 10% disease penetrance that permits Mtb propagation. This selective pressure over many millennia would progressively remove genotypes that lead to high susceptibility to TB, but equally tend to select against individuals with complete resistance to Mtb colonisation (**Figure 2** [↗](#)). This could explain why consistent genetic traits for susceptibility or resistance to TB have been hard to identify (39 [↗](#)). More recent mass infection events, such as the smallpox epidemics that killed approximately 25% of the population (67 [↗](#)), may have further favoured individuals immunologically trained by Mtb. Likewise, successive waves of plague killed approximately 25% of the population, at the end of the Roman empire and then in the European middle ages, adding to selective pressure from endemic infections (68 [↗](#), 69 [↗](#)). If humans have been selected to be permissive to Mtb colonisation but broadly resistant to disease, it suggests the development of active disease must be a relatively unusual event in a subset of outlier individuals. In some sense, we could be regarded as having a symbiotic relationship with Mtb, with disease representing a necessary evil, caused by an imbalance in the predominantly stable host pathogen interaction (70 [↗](#), 71 [↗](#)).

The host-pathogen interaction at a cellular level

The early histological era described the wide range of human TB lesions and granuloma types, and identified TB as a disease characterised by spatial organisation (72 [↗](#)). Classically, the granuloma has been proposed to be where the outcome of the host-pathogen interaction is determined (73 [↗](#)). Recent methodological advances are permitting much greater dissection of events and further highlight the importance of spatial organisation within the granuloma (74 [↗](#)–76 [↗](#)). However, just as the early X-ray era showed some lesions progressing and some regressing, these studies demonstrate the great heterogeneity between granuloma types. Studies in the non-human primate have allowed investigation into features of progressing and controlling granulomas, identifying potential correlates of immune control (77 [↗](#)), but even this model only partially recapitulates human disease.

Furthermore, the recent spatial studies highlight the complexity of cellular players, including the established fulcrum of macrophages and T cells, but additionally the importance of B cells, neutrophils and fibroblasts. For example, fibroblasts are emerging as important immune regulators in other lung diseases (78 [↗](#)), and so it seems highly likely that they play an active role in TB-related inflammation. Fibroblast zonation can lead to feed-forward inflammatory loops and so may propagate disease (79 [↗](#)). Consequently, the full spectrum of cell types in Mtb-infected lesions needs to be considered. Given that multiple immunological routes can lead to active TB, it seems unlikely that a single cellular component will fully explain the balance between Mtb containment and progression to active disease. However, the majority of studies continue to look for a single protective immune mechanism (41 [↗](#), 77 [↗](#), 80 [↗](#)–82 [↗](#)).

Considering the clinical and experimental evidence, immunological failure is likely to be a multistep process, whereby either one large deficit, or numerous small imbalances, can lead to progression and disease (**Figure 3** [↗](#)). Mtb and humans interact over many years, as the pathogen is difficult to eradicate due to its highly evolved survival mechanisms (83 [↗](#)), and therefore in those individuals in whom it survives, there is a long period where it can escape host control.

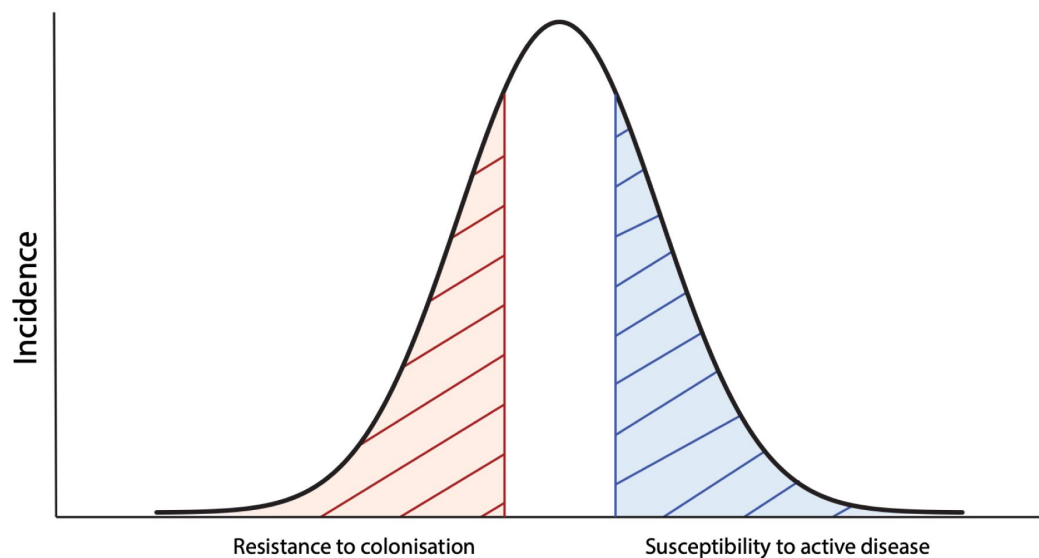


Figure 2

Selection pressure of prolonged co-evolution favours individuals permissive to colonisation but resistant to active disease.

Over millennia, *Mtb* circulation in society will remove genetic traits resulting in high susceptibility to active TB infection. Perhaps less intuitively, if *Mtb* generates trained immunity that protects against other fatal infections, individuals highly resistant to *Mtb* colonisation will also be selected against. The resulting population would then reflect modern humans; susceptible to initial *Mtb* infection but generally resistant to TB disease.

Potentially these different paths may ultimately cross or converge; if so, understanding the key nodes will allow more broadly effective treatments to emerge. For example, lung extracellular matrix degradation could be regarded as a final common pathway (84 [↗](#)), but, again, this may result from different collagenases including macrophage-derived MMP-1 or neutrophil-derived MMP-8 (85 [↗](#), 86 [↗](#)).

Emerging methodologies and the challenge of data analysis

The recent adoption of “omic” methodologies, including single cell transcriptomics, spatial transcriptomics, and proteomics, offers unprecedented opportunities to dissect the mechanisms of human TB pathogenesis and accelerate the development of effective interventions. These approaches generate large-scale, complex datasets that capture the heterogeneity of TB lesions. However, this complexity also presents significant analytical challenges. Standard computational pipelines, if not adapted to account for the biological and technical variability in these data, are unlikely to deliver robust or reproducible insights. A major obstacle is the integration of data from multiple studies and platforms, which can differ both within a single omic layer (horizontal integration) and across multiple omic layers (vertical integration) (87 [↗](#)). Such integration risks data loss and inconsistent results, especially when data are not harmonised.

The prolonged co-evolution between host and pathogen has resulted in multiple routes to active TB, significantly adding to the biological heterogeneity that complicates the analysis and interpretation of multi-omic data. Addressing this complexity requires well-annotated clinical cohorts that capture the full spectrum of TB heterogeneity, ideally with longitudinal outcome data rather than single time-point snapshots. To accommodate the diversity of disease pathways, novel bioinformatic approaches are needed that move beyond the assumption of a single, unifying route to active TB.

The power of multi-omic approaches to reveal complex molecular relationships depends on both the quality of the omic data and the fit between experimental design and data integration strategies. For example, correlation-based integration requires matched samples across omics, sufficient sample numbers, and comparable variance structures. These requirements are often overlooked, leading to insufficient power, noisy data, and unrealistic integration (88 [↗](#)). Each omic technology brings its own challenges. Signal-to-noise ratios differ across modalities, and appropriate algorithms are needed to estimate sample size and power for each. Notably, using the same sample numbers across modalities does not ensure comparable statistical power; achieving equal power can require unbalanced sample sizes (89 [↗](#)). Other recognised challenges include the interpretation and validation of multi-omic models, standardised annotation, and data sharing (89 [↗](#)). Recent developments include webbased, user-friendly tools that enable both knowledge- and data-driven multi-omic integration (90 [↗](#)). Importantly, a wider use of artificial intelligence (AI) is transforming omics data analysis by providing robust methods to interpret complex biological datasets (91 [↗](#)). Machine learning and deep learning techniques are now routinely applied to DNA, transcriptomic, proteomic, and metabolomic data, enabling more integrated and comprehensive analyses (91 [↗](#)). In proteomics, for example, AI-driven approaches have enhanced peptide measurement predictions and accelerated biomarker discovery, often outperforming conventional assays (92 [↗](#)). To improve interpretability, explainable AI (XAI) methods are increasingly employed, with feature relevance mapping and visual explanations emerging as preferred post hoc strategies (93 [↗](#)). However, despite these developments, significant challenges remain in implementing XAI. Further research is needed to overcome these barriers and unlock the full translational potential of AI in omics analysis (92 [↗](#), 93 [↗](#)). Ultimately, the utility of these innovations will depend on the functional validation of the resulting models.

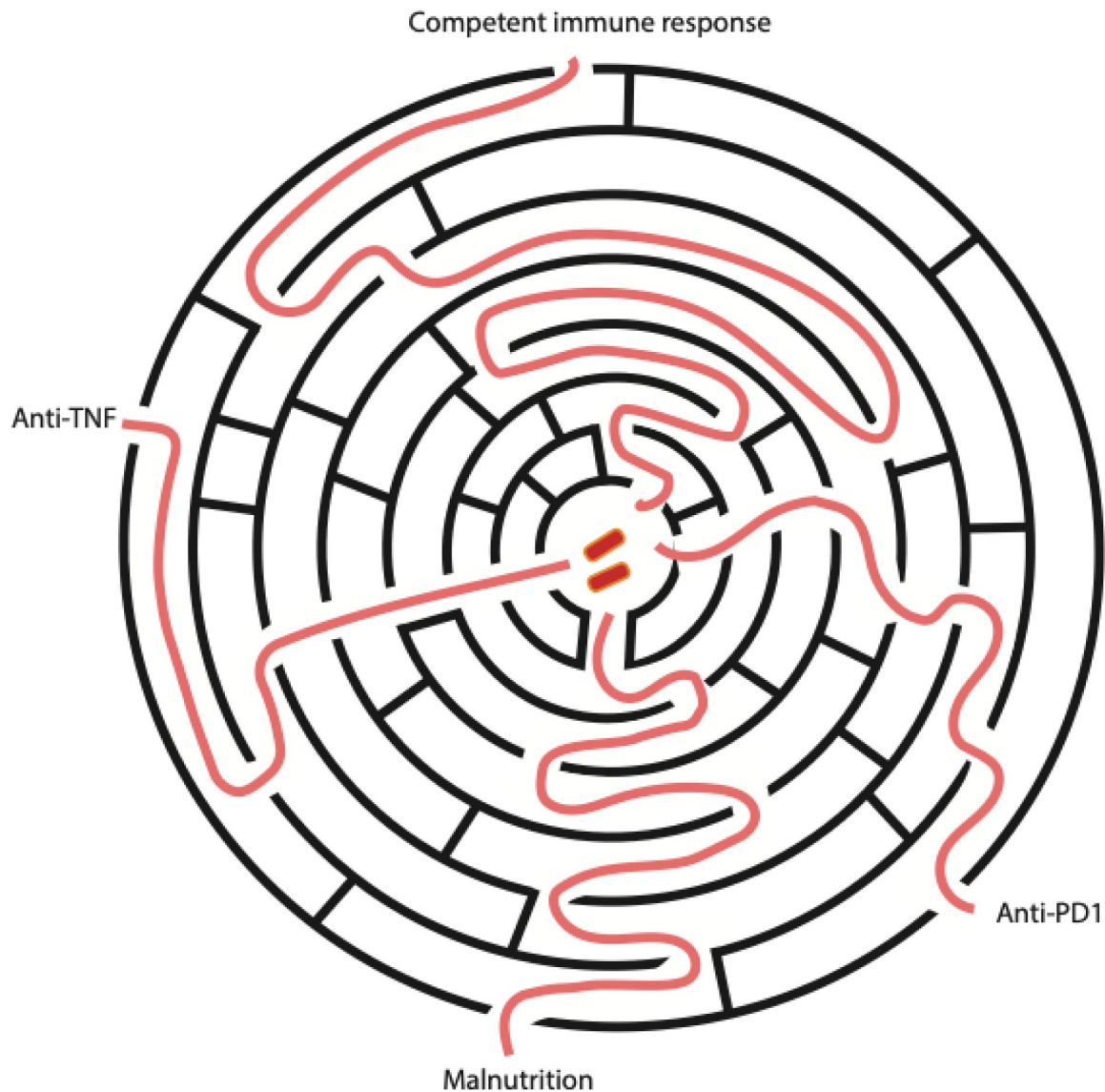


Figure 3

Escaping the human immune response from the pathogen's perspective.

As control of Mtb requires a co-ordinated host response, there are multiple routes that Mtb can escape by. A major immune disturbance, such as TNF- α or PD-1 inhibition, gives a relatively direct route of exit. However, most individuals develop TB due to a series of less apparent immune events and no obvious global immune disturbance.

Given the complexity of human TB, careful study design and unbiased integration methods that accommodate data limitations are essential. Spatial context is particularly important, as TB pathology involves three-dimensional immune responses. Extracellular matrix remodelling is a hallmark of TB granulomas [81], and the matrix regulates host cell biology [84]. Consequently, multiple data inputs such as matrix composition and organisation may be needed alongside host and Mtb transcriptomic data. Ultimately, local cellular events must be modelled at the tissue level, providing a second layer of computational complexity [85].

Addressing these challenges will require substantial investment in analytical capacity. Multi-omics is already transforming our understanding of disease heterogeneity, facilitating the identification of previously unrecognized subgroups, refining prognostic and therapeutic approaches, and providing deeper mechanistic insights. This strategy has successfully stratified rare tumours (94 [↗](#)), profiled healthy populations (95 [↗](#)), and enabled health screening to reveal previously hidden disease and risk subgroups (96 [↗](#)), supporting the shift toward precision medicine. As human disease processes are rarely uniform, advances in multi-omic study design and analysis for TB will likely benefit a broad range of conditions.

Implications of diverse disease pathways for new TB interventions

Ultimately, the complexity of human TB underlies our inability to deploy a transformative intervention, and so global mortality remains depressingly high. The human-Mtb interaction is so closely co-evolved that experimental findings need to be interpreted in light of human disease phenomena. Multiomic studies in TB are currently being undertaken on small numbers due to cost and challenges in obtaining appropriate clinical samples, in specific regions, and so are unlikely to reflect the global heterogeneity. Therefore, results need to be interpreted with caution and wider studies will be needed to confirm the generalisability of findings. A critical aspect will be carefully curated and fully accessible metadata, so that as the body of omic datasets on human TB increase, they can be accurately integrated into wider analyses. Recurrent mining of these datasets is likely to be fundamental to understanding the breadth of TB pathogenesis. Missing metadata makes interpretation difficult, and in worst cases misleading. In addition, innovative computational approaches will be required whereby the analysis specifically accommodates multiple immune pathways to disease.

Given the toll that TB takes in the poorest parts of the world, we have a moral imperative to end the epidemic (97 [↗](#)). To achieve this, we must acknowledge the complexity of human TB that has resulted from our prolonged co-evolution with the pathogen and the selective pressure of persistent Mtb exposure over millennia. Success depends on integrating the full spectrum of TB disease into our bioinformatic analyses, and ultimately understanding TB's complexity can then inform logical interventions. If we seek a single unified explanation of TB disease, this seems to be unlikely to be successful.

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References

1. Kaufmann SH (2003) **A short history of Robert Koch's fight against tuberculosis: Those who do not remember the past are condemned to repeat it** *Tuberculosis (Edinb)* **83**:86–90 [PubMed](#) | [Google Scholar](#)
2. O'Garra A, Redford PS, McNab FW, Bloom CI, Wilkinson RJ, Berry MP (2013) **The immune response in tuberculosis** *Annu Rev Immunol* **31**:475–527 <https://doi.org/10.1146/annurev-immunol-032712-095939> | [PubMed](#) | [Google Scholar](#)
3. Trajman A, Campbell JR, Kunor T, Ruslami R, Amanullah F, Behr MA, Menzies D (2025) **Tuberculosis** *Lancet* **405**:850–66 [https://doi.org/10.1016/S0140-6736\(24\)02479-6](https://doi.org/10.1016/S0140-6736(24)02479-6) | [PubMed](#) | [Google Scholar](#)
4. Mangtani P, Abubakar I, Ariti C, Beynon R, Pimpin L, Fine PE, et al. (2014) **Protection by BCG Vaccine Against Tuberculosis: A Systematic Review of Randomized Controlled Trials** *Clin Infect Dis* **58**:470–80 <https://doi.org/10.1093/cid/cit790> | [PubMed](#) | [Google Scholar](#)
5. Trunz BB, Fine P, Dye C (2006) **Effect of BCG vaccination on childhood tuberculous meningitis and miliary tuberculosis worldwide: a meta-analysis and assessment of cost-effectiveness** *Lancet* **367**:1173–80 [https://doi.org/10.1016/S0140-6736\(06\)68507-3](https://doi.org/10.1016/S0140-6736(06)68507-3) | [PubMed](#) | [Google Scholar](#)
6. Tait DR, Hatherill M, Van Der Meeren O, Ginsberg AM, Van Brakel E, Salaun B, et al. (2019) **Final Analysis of a Trial of M72/AS01E Vaccine to Prevent Tuberculosis** *N Engl J Med* **381**:2429–39 <https://doi.org/10.1056/NEJMoa1909953> | [PubMed](#) | [Google Scholar](#)
7. Borges AH, Russell M, Tait D, Scriba TJ, Nemes E, Skallerup P, et al. (2025) **Immunogenicity, safety, and efficacy of the vaccine H56:IC31 in reducing the rate of tuberculosis disease recurrence in HIV-negative adults successfully treated for drug-susceptible pulmonary tuberculosis: a double-blind, randomised, placebo-controlled, phase 2b trial** *Lancet Infect Dis* [https://doi.org/10.1016/S1473-3099\(24\)00814-4](https://doi.org/10.1016/S1473-3099(24)00814-4) | [PubMed](#) | [Google Scholar](#)
8. Schmidt AC, Fairlie L, Hellstrom E, Luabeya Kany Kany A, Middelkoop K, Naidoo K, et al. (2025) **BCG Revaccination for the Prevention of Mycobacterium tuberculosis Infection** *N Engl J Med* **392**:1789–800 <https://doi.org/10.1056/NEJMoa2412381> | [PubMed](#) | [Google Scholar](#)
9. Brites D, Gagneux S (2015) **Co-evolution of Mycobacterium tuberculosis and Homo sapiens** *Immunol Rev* **264**:6–24 <https://doi.org/10.1111/imr.12264> | [PubMed](#) | [Google Scholar](#)
10. Dutour O (2023) **The paleopathology and paleoepidemiology of Upper paleolithic tuberculosis: Review of evidence and hypotheses** *Tuberculosis (Edinb)* **143S**:102348 <https://doi.org/10.1016/j.tube.2023.102348> | [PubMed](#) | [Google Scholar](#)
11. Gagneux S (2012) **Host-pathogen coevolution in human tuberculosis** *Philos Trans R Soc Lond B Biol Sci* **367**:850–9 <https://doi.org/10.1098/rstb.2011.0316> | [PubMed](#) | [Google Scholar](#)

12. Dubos R, Dubos J (1987) **The white plague : tuberculosis, man, and society** New Brunswick: London: Rutgers University Press [Google Scholar](#)
13. Ghon A, Barty King D. (1916) **The Primary Lung Focus of Tuberculosis in Children** London: J. & A. Churchill [Google Scholar](#)
14. Elkington PT, Friedland JS (2015) **Permutations of time and place in tuberculosis** *Lancet Infect Dis* **15**:1357–60 [https://doi.org/10.1016/S1473-3099\(15\)00135-8](https://doi.org/10.1016/S1473-3099(15)00135-8) | PubMed | [Google Scholar](#)
15. Kwan CK, Ernst JD (2011) **HIV and tuberculosis: a deadly human syndemic** *Clin Microbiol Rev* **24**:351–76 <https://doi.org/10.1128/CMR.00042-10> | PubMed | [Google Scholar](#)
16. Keane J, Gershon S, Wise RP, Mirabile-Levens E, Kasznica J, Schwieterman WD, et al. (2001) **Tuberculosis associated with infliximab, a tumor necrosis factor alpha-neutralizing agent** *N Engl J Med* **345**:1098–104 <https://doi.org/10.1056/NEJMoa011110> | PubMed | [Google Scholar](#)
17. Jouanguy E, Altare F, Lamhamedi S, Revy P, Emile JF, Newport M, et al. (1996) **Interferon-gamma-receptor deficiency in an infant with fatal bacille Calmette-Guerin infection** *N Engl J Med* **335**:1956–61 [PubMed](#) | [Google Scholar](#)
18. Dupuis S, Dargemont C, Fieschi C, Thomassin N, Rosenzweig S, Harris J, et al. (2001) **Impairment of mycobacterial but not viral immunity by a germline human STAT1 mutation** *Science* **293**:300–3 <https://doi.org/10.1126/science.1061154> | PubMed | [Google Scholar](#)
19. Arias AA, Neehus AL, Ogishi M, Meynier V, Krebs A, Lazarov T, et al. (2024) **Tuberculosis in otherwise healthy adults with inherited TNF deficiency** *Nature* **633**:417–25 <https://doi.org/10.1038/s41586-024-07866-3> | PubMed | [Google Scholar](#)
20. Dheda K, Barry CE, Maartens G (2016) **Tuberculosis** *Lancet* **387**:1211–26 [https://doi.org/10.1016/S0140-6736\(15\)00151-8](https://doi.org/10.1016/S0140-6736(15)00151-8) | PubMed | [Google Scholar](#)
21. Bhargava A, Bhargava M, Meher A, Benedetti A, Velayutham B, Sai Teja G, et al. (2023) **Nutritional supplementation to prevent tuberculosis incidence in household contacts of patients with pulmonary tuberculosis in India (RATIONS): a field-based, open-label, cluster-randomised, controlled trial** *Lancet* **402**:627–40 [https://doi.org/10.1016/S0140-6736\(23\)01231-X](https://doi.org/10.1016/S0140-6736(23)01231-X) | PubMed | [Google Scholar](#)
22. Comstock GW, Livesay VT, Woolpert SF (1974) **The prognosis of a positive tuberculin reaction in childhood and adolescence** *American journal of epidemiology* **99**:131–8 [PubMed](#) | [Google Scholar](#)
23. Andrews JR, Nemes E, Tameris M, Landry BS, Mahomed H, McClain JB, et al. (2017) **Serial QuantiFERON testing and tuberculosis disease risk among young children: an observational cohort study** *The lancet Respiratory medicine* **5**:282–90 [https://doi.org/10.1016/S2213-2600\(17\)30060-7](https://doi.org/10.1016/S2213-2600(17)30060-7) | PubMed | [Google Scholar](#)
24. Ledesma JR, Ma J, Zheng P, Ross JM, Vos T, Kyu HH (2021) **Interferon-gamma release assay levels and risk of progression to active tuberculosis: a systematic review and dose-response meta-regression analysis** *BMC infectious diseases* **21**:467 <https://doi.org/10.1186/s12879-021-06141-4> | PubMed | [Google Scholar](#)
25. Horton KC, MacPherson P, Houben RM, White RG, Corbett EL (2016) **Sex Differences in Tuberculosis Burden and Notifications in Low-and Middle-Income Countries: A**

Systematic Review and Meta-analysis *PLoS Med* **13**:e1002119 <https://doi.org/10.1371/journal.pmed.1002119> | PubMed | Google Scholar

26. Tezera LB, Mansour S, Elkington P (2020) **Reconsidering the Optimal Immune Response to *Mycobacterium tuberculosis*** *Am J Respir Crit Care Med* **201**:407–13 <https://doi.org/10.1164/rccm.201908-1506PP> | PubMed | Google Scholar
27. Elkington PT, Bateman AC, Thomas GJ, Ottensmeier CH (2018) **Implications of Tuberculosis Reactivation after Immune Checkpoint Inhibition** *Am J Respir Crit Care Med* **198**:1451–3 <https://doi.org/10.1164/rccm.201807-1250LE> | PubMed | Google Scholar
28. Lazar-Molnar E, Chen B, Sweeney KA, Wang EJ, Liu W, Lin J, et al. (2010) **Programmed death-1 (PD-1)-deficient mice are extraordinarily sensitive to tuberculosis** *Proc Natl Acad Sci U S A* **107**:13402–7 <https://doi.org/10.1073/pnas.1007394107> | PubMed | Google Scholar
29. Barber DL, Mayer-Barber KD, Feng CG, Sharpe AH, Sher A (2011) **CD4 T cells promote rather than control tuberculosis in the absence of PD-1-mediated inhibition** *J Immunol* **186**:1598–607 <https://doi.org/10.4049/jimmunol.1003304> | PubMed | Google Scholar
30. Kauffman KD, Sakai S, Lora NE, Namasivayam S, Baker PJ, Kamenyeva O, et al. (2021) **PD-1 blockade exacerbates *Mycobacterium tuberculosis* infection in rhesus macaques** *Sci Immunol* **6** <https://doi.org/10.1126/sciimmunol.abf3861> | PubMed | Google Scholar
31. Tezera LB, Bielecka MK, Ogongo P, Walker NF, Ellis M, Garay-Baquero DJ, et al. (2020) **Anti-PD-1 immunotherapy leads to tuberculosis reactivation via dysregulation of TNF- α** *eLife* **9** <https://doi.org/10.7554/eLife.52668> | PubMed | Google Scholar
32. Liu K, Wang D, Yao C, Qiao M, Li Q, Ren W, et al. (2022) **Increased Tuberculosis Incidence Due to Immunotherapy Based on PD-1 and PD-L1 Blockade: A Systematic Review and Meta-Analysis** *Frontiers in immunology* **13**:727220 <https://doi.org/10.3389/fimmu.2022.727220> | PubMed | Google Scholar
33. Zhu J, He Z, Liang D, Yu X, Qiu K, Wu J (2022) **Pulmonary tuberculosis associated with immune checkpoint inhibitors: a pharmacovigilance study** *Thorax* <https://doi.org/10.1136/thoraxjnl-2021-217575> | PubMed | Google Scholar
34. Eckold C, Kumar V, Weiner J, Alisjahbana B, Riza AL, Ronacher K, et al. (2021) **Impact of Intermediate Hyperglycemia and Diabetes on Immune Dysfunction in Tuberculosis** *Clin Infect Dis* **72**:69–78 <https://doi.org/10.1093/cid/ciaa751> | PubMed | Google Scholar
35. Behr MA, Edelstein PH, Ramakrishnan L (2024) **Rethinking the burden of latent tuberculosis to reprioritize research** *Nat Microbiol* **9**:1157–8 <https://doi.org/10.1038/s41564-024-01683-0> | PubMed | Google Scholar
36. Houben RM, Dodd PJ (2016) **The Global Burden of Latent Tuberculosis Infection: A Re-estimation Using Mathematical Modelling** *PLoS Med* **13**:e1002152 <https://doi.org/10.1371/journal.pmed.1002152> | PubMed | Google Scholar
37. Schurz H, Naranbhai V, Yates TA, Gilchrist JJ, Parks T, Dodd PJ, et al. (2024) **Multi-ancestry meta-analysis of host genetic susceptibility to tuberculosis identifies shared genetic architecture** *eLife* **13** <https://doi.org/10.7554/eLife.84394> | PubMed | Google Scholar
38. Bellamy R, Ruwende C, Corrah T, McAdam KP, Whittle HC, Hill AV (1998) **Variations in the *NRAMP1* gene and susceptibility to tuberculosis in West Africans** *N Engl J Med* **338**:640–

- 4 <https://doi.org/10.1056/NEJM199803053381002> | PubMed | Google Scholar
39. Abel L, El-Baghdadi J, Bousfiha AA, Casanova JL, Schurr E (2014) **Human genetics of tuberculosis: a long and winding road** *Philos Trans R Soc Lond B Biol Sci* **369**:20130428 <https://doi.org/10.1098/rstb.2013.0428> | PubMed | Google Scholar
40. Cross DL, Layton ED, Yu KK, Smith MT, Aguilar MS, Li S, et al. (2024) **MR1-restricted T cell clonotypes are associated with “resistance” to Mycobacterium tuberculosis infection** *JCI Insight* **9** <https://doi.org/10.1172/jci.insight.166505> | PubMed | Google Scholar
41. Sun M, Phan JM, Kieswetter NS, Huang H, Yu KKQ, Smith MT, et al. (2024) **Specific CD4(+) T cell phenotypes associate with bacterial control in people who ‘resist’ infection with Mycobacterium tuberculosis** *Nat Immunol* **25**:1411–21 <https://doi.org/10.1038/s41590-024-01897-8> | PubMed | Google Scholar
42. Dallmann-Sauer M, Fava VM, Malherbe ST, MacDonald CE, Orlova M, Kroon EE, et al. (2025) **Mycobacterium tuberculosis resisters despite HIV exhibit activated T cells and macrophages in their pulmonary alveoli** *J Clin Invest* <https://doi.org/10.1172/JCI188016> | PubMed | Google Scholar
43. Simmons JD, Dill-McFarland KA, Stein CM, Van PT, Chihota V, Ntshiqha T, et al. (2022) **Monocyte Transcriptional Responses to Mycobacterium tuberculosis Associate with Resistance to Tuberculin Skin Test and Interferon Gamma Release Assay Conversion** *mSphere* **9**:e0015922 <https://doi.org/10.1128/msphere.00159-22> | PubMed | Google Scholar
44. Lu LL, Smith MT, Yu KKQ, Luedemann C, Suscovich TJ, Grace PS, et al. (2019) **IFN-gamma-independent immune markers of Mycobacterium tuberculosis exposure** *Nat Med* **25**:977–87 <https://doi.org/10.1038/s41591-019-0441-3> | PubMed | Google Scholar
45. Verrall AJ, Schneider M, Alisjahbana B, Apriani L, van Laarhoven A, Koeken V, et al. (2020) **Early Clearance of Mycobacterium tuberculosis Is Associated With Increased Innate Immune Responses** *J Infect Dis* **221**:1342–50 <https://doi.org/10.1093/infdis/jiz147> | PubMed | Google Scholar
46. Smith CM, Baker RE, Proulx MK, Mishra BB, Long JE, Park SW, et al. (2022) **Host pathogen genetic interactions underlie tuberculosis susceptibility in genetically diverse mice** *eLife* **11** <https://doi.org/10.7554/eLife.74419> | PubMed | Google Scholar
47. Yoder MA, Lamichhane G, Bishai WR (2004) **Cavitary pulmonary tuberculosis: The Holy Grail of disease transmission** *Current Sci* **86**:74–81 <https://doi.org/10.1006/csc.2004.3381> | Google Scholar
48. Elkington P, Lerm M, Kapoor N, Mahon R, Pienaar E, Huh D, et al. (2019) **In Vitro Granuloma Models of Tuberculosis: Potential and Challenges** *J Infect Dis* **219**:1858–66 <https://doi.org/10.1093/infdis/jiz020> | PubMed | Google Scholar
49. Gros P, Casanova JL (2023) **Reconciling Mouse and Human Immunology at the Altar of Genetics** *Annu Rev Immunol* **41**:39–71 <https://doi.org/10.1146/annurev-immunol-101721-065201> | PubMed | Google Scholar
50. Young D (2009) **Animal models of tuberculosis** *Eur J Immunol* **39**:2011–4 <https://doi.org/10.1002/eji.200939542> | PubMed | Google Scholar
51. DiNardo AR, Nishiguchi T, Grimm SL, Schlesinger LS, Graviss EA, Cirillo JD, et al. (2021) **Tuberculosis endotypes to guide stratified host-directed therapy** *Med (N Y)* **2**:217–32 <https://doi.org/10.1016/j.med.2021.05.001> | PubMed | Google Scholar

[//doi.org/10.1016/j.medj.2020.11.003](https://doi.org/10.1016/j.medj.2020.11.003) | PubMed | Google Scholar

52. Koleske BN, Jacobs WR, Bishai WR (2023) **The Mycobacterium tuberculosis genome at 25 years: lessons and lingering questions** *J Clin Invest* **133** <https://doi.org/10.1172/JCI173156> | PubMed | Google Scholar
53. Goig GA, Windels EM, Loiseau C, Stritt C, Biru L, Borrell S, et al. (2025) **Ecology, global diversity and evolutionary mechanisms in the Mycobacterium tuberculosis complex** *Nat Rev Microbiol* <https://doi.org/10.1038/s41579-025-01159-w> | PubMed | Google Scholar
54. Yenew B, Ghodousi A, Diriba G, Tesfaye E, Cabibbe AM, Amare M, et al. (2023) **A smooth tubercle bacillus from Ethiopia phylogenetically close to the Mycobacterium tuberculosis complex** *Nature communications* **14**:7519 <https://doi.org/10.1038/s41467-023-42755-9> | PubMed | Google Scholar
55. Garnier T, Eiglmeier K, Camus JC, Medina N, Mansoor H, Pryor M, et al. (2003) **The complete genome sequence of Mycobacterium bovis** *Proc Natl Acad Sci U S A* **100**:7877–82 <https://doi.org/10.1073/pnas.1130426100> | PubMed | Google Scholar
56. Comas I, Chakravartti J, Small PM, Galagan J, Niemann S, Kremer K, et al. (2010) **Human T cell epitopes of Mycobacterium tuberculosis are evolutionarily hyperconserved** *Nat Genet* **42**:498–503 <https://doi.org/10.1038/ng.590> | PubMed | Google Scholar
57. Nathan C (2023) **Mycobacterium tuberculosis as teacher** *Nat Microbiol* <https://doi.org/10.1038/s41564-023-01454-3> | PubMed | Google Scholar
58. Khan N, Downey J, Sanz J, Kaufmann E, Blankenhaus B, Pacis A, et al. (2020) **M. tuberculosis Reprograms Hematopoietic Stem Cells to Limit Myelopoiesis and Impair Trained Immunity** *Cell* **183**:752–70 e22 <https://doi.org/10.1016/j.cell.2020.09.062> | PubMed | Google Scholar
59. Kaufmann E, Sanz J, Dunn JL, Khan N, Mendonca LE, Pacis A, et al. (2018) **BCG Educates Hematopoietic Stem Cells to Generate Protective Innate Immunity against Tuberculosis** *Cell* **172**:176–90 e19 <https://doi.org/10.1016/j.cell.2017.12.031> | PubMed | Google Scholar
60. Jurczak M, Druszczyńska M (2025) **Beyond Tuberculosis: The Surprising Immunological Benefits of the Bacillus Calmette-Guerin (BCG) Vaccine in Infectious, Auto-Immune, and Inflammatory Diseases** *Pathogens* **14** <https://doi.org/10.3390/pathogens14020196> | PubMed | Google Scholar
61. Moorlag S, Arts RJW, van Crevel R, Netea MG (2019) **Non-specific effects of BCG vaccine on viral infections** *Clin Microbiol Infect* **25**:1473–8 <https://doi.org/10.1016/j.cmi.2019.04.020> | PubMed | Google Scholar
62. Stensballe LG, Nante E, Jensen IP, Kofoed PE, Poulsen A, Jensen H, et al. (2005) **Acute lower respiratory tract infections and respiratory syncytial virus in infants in Guinea-Bissau: a beneficial effect of BCG vaccination for girls community based case-control study** *Vaccine* **23**:1251–7 <https://doi.org/10.1016/j.vaccine.2004.09.006> | PubMed | Google Scholar
63. Arts RJW, Moorlag S, Novakovic B, Li Y, Wang SY, Oosting M, et al. (2018) **BCG Vaccination Protects against Experimental Viral Infection in Humans through the Induction of**

Cytokines Associated with Trained Immunity *Cell Host Microbe* **23**:89–100 e5 <https://doi.org/10.1016/j.chom.2017.12.010> | PubMed | Google Scholar

64. Rieckmann A, Villumsen M, Sorup S, Haugaard LK, Ravn H, Roth A, et al. (2017) **Vaccinations against smallpox and tuberculosis are associated with better long-term survival: a Danish case-cohort study 1971-2010** *International journal of epidemiology* **46**:695–705 <https://doi.org/10.1093/ije/dyw120> | PubMed | Google Scholar
65. Vallini L, Zampieri C, Shoaee MJ, Bortolini E, Marciani G, Aneli S, et al. (2024) **The Persian plateau served as hub for Homo sapiens after the main out of Africa dispersal** *Nature communications* **15**:1882 <https://doi.org/10.1038/s41467-024-46161-7> | PubMed | Google Scholar
66. Bergstrom A, Stringer C, Hajdinjak M, Scerri EML, Skoglund P (2021) **Origins of modern human ancestry** *Nature* **590**:229–37 <https://doi.org/10.1038/s41586-021-03244-5> | PubMed | Google Scholar
67. Dobson AP, Carper ER (1996) **Infectious diseases and human population history** *Bioscience* **46**:115–26 [Google Scholar](#)
68. Little LK (2007) **Plague and the end of antiquity : the pandemic of 541-750** New York: Cambridge University Press :360 [Google Scholar](#)
69. Benedictow OJ (2004) **The Black Death, 1346-1353 : the complete history** Woodbridge, Suffolk, UK ; Rochester, N.Y., USA: Boydell Press :433 [Google Scholar](#)
70. Divangahi M, Khan N, Kaufmann E (2018) **Beyond Killing Mycobacterium tuberculosis: Disease Tolerance** *Frontiers in immunology* **9**:2976 <https://doi.org/10.3389/fimmu.2018.02976> | PubMed | Google Scholar
71. Olive AJ, Sasseti CM (2018) **Tolerating the Unwelcome Guest; How the Host Withstands Persistent Mycobacterium tuberculosis** *Frontiers in immunology* **9**:2094 <https://doi.org/10.3389/fimmu.2018.02094> | PubMed | Google Scholar
72. Hunter RL (2016) **Tuberculosis as a three-act play: A new paradigm for the pathogenesis of pulmonary tuberculosis** *Tuberculosis (Edinb)* **97**:8–17 <https://doi.org/10.1016/j.tube.2015.11.010> | PubMed | Google Scholar
73. Pagan AJ, Ramakrishnan L (2018) **The Formation and Function of Granulomas** *Annu Rev Immunol* **36**:639–65 <https://doi.org/10.1146/annurev-immunol-032712-100022> | PubMed | Google Scholar
74. Sawyer AJ, Patrick E, Edwards J, Wilmott JS, Fielder T, Yang Q, et al. (2023) **Spatial mapping reveals granuloma diversity and histopathological superstructure in human tuberculosis** *J Exp Med* **220** <https://doi.org/10.1084/jem.20221392> | PubMed | Google Scholar
75. McCaffrey EF, Donato M, Keren L, Chen Z, Delmastro A, Fitzpatrick MB, et al. (2022) **The immunoregulatory landscape of human tuberculosis granulomas** *Nat Immunol* <https://doi.org/10.1038/s41590-021-01121-x> | PubMed | Google Scholar

76. Marakalala MJ, Raju RM, Sharma K, Zhang YJ, Eugenin EA, Prideaux B, et al. (2016) **Inflammatory signaling in human tuberculosis granulomas is spatially organized** *Nat Med* **22**:531–8 <https://doi.org/10.1038/nm.4073> | PubMed | Google Scholar
77. Gideon HP, Hughes TK, Tzouanas CN, Wadsworth MH, Tu AA, Gierahn TM, et al. (2022) **Multimodal profiling of lung granulomas in macaques reveals cellular correlates of tuberculosis control** *Immunity* **55**:827–46 e10 <https://doi.org/10.1016/j.immuni.2022.04.004> | PubMed | Google Scholar
78. Ghonim MA, Boyd DF, Flerlage T, Thomas PG (2023) **Pulmonary inflammation and fibroblast immunoregulation: from bench to bedside** *J Clin Invest* **133** <https://doi.org/10.1172/JCI170499> | PubMed | Google Scholar
79. Davidson S, Coles M, Thomas T, Kollias G, Ludewig B, Turley S, et al. (2021) **Fibroblasts as immune regulators in infection, inflammation and cancer** *Nat Rev Immunol* **21**:704–17 <https://doi.org/10.1038/s41577-021-00540-z> | PubMed | Google Scholar
80. Winchell CG, Nyquist SK, Chao MC, Maiello P, Myers AJ, Hopkins F, et al. (2023) **CD8+ lymphocytes are critical for early control of tuberculosis in macaques** *J Exp Med* **220** <https://doi.org/10.1084/jem.20230707> | PubMed | Google Scholar
81. Swanson RV, Gupta A, Foreman TW, Lu L, Choreno-Parra JA, Mbandi SK, et al. (2023) **Antigen-specific B cells direct T follicular-like helper cells into lymphoid follicles to mediate Mycobacterium tuberculosis control** *Nat Immunol* <https://doi.org/10.1038/s41590-023-01476-3> | PubMed | Google Scholar
82. Proulx MK, Wiggins CD, Reames CJ, Wu C, Kiritsy MC, Xu P, et al. (2025) **Noncanonical T cell responses are associated with protection from tuberculosis in mice and humans** *J Exp Med* **222** <https://doi.org/10.1084/jem.20241760> | PubMed | Google Scholar
83. Russell DG (2011) **Mycobacterium tuberculosis and the intimate discourse of a chronic infection** *Immunol Rev* **240**:252–68 <https://doi.org/10.1111/j.1600-065X.2010.00984.x> | PubMed | Google Scholar
84. Elkington P, Polak ME, Reichmann MT, Leslie A (2022) **Understanding the tuberculosis granuloma: the matrix revolutions** *Trends Mol Med* **28**:143–54 <https://doi.org/10.1016/j.molmed.2021.11.004> | PubMed | Google Scholar
85. Elkington P, Shiomi T, Breen R, Nuttall RK, Ugarte-Gil CA, Walker NF, et al. (2011) **MMP-1 drives immunopathology in human tuberculosis and transgenic mice** *J Clin Invest* **121**:1827–33 <https://doi.org/10.1172/JCI45666> | PubMed | Google Scholar
86. Ong CW, Elkington PT, Brilha S, Ugarte-Gil C, Tome-Esteban MT, Tezera LB, et al. (2015) **Neutrophil-Derived MMP-8 Drives AMPK-Dependent Matrix Destruction in Human Pulmonary Tuberculosis** *PLoS Pathog* **11**:e1004917 <https://doi.org/10.1371/journal.ppat.1004917> | PubMed | Google Scholar
87. Zheng Y, Liu Y, Yang J, Dong L, Zhang R, Tian S, et al. (2024) **Multi-omics data integration using ratio-based quantitative profiling with Quartet reference materials** *Nat Biotechnol* **42**:1133–49 <https://doi.org/10.1038/s41587-023-01934-1> | PubMed | Google Scholar
88. Tarazona S, Balzano-Nogueira L, Gomez-Cabrero D, Schmidt A, Imhof A, Hankemeier T, et al. (2020) **Harmonization of quality metrics and power calculation in multi-omic studies**

Nature communications **11**:3092 <https://doi.org/10.1038/s41467-020-16937-8> | PubMed | Google Scholar

89. Tarazona S, Arzalluz-Luque A, Conesa A (2021) **Undisclosed, unmet and neglected challenges in multi-omics studies** *Nat Comput Sci* **1**:395–402 <https://doi.org/10.1038/s43588-021-00086-z> | PubMed | Google Scholar
90. Ewald JD, Zhou G, Lu Y, Kolic J, Ellis C, Johnson JD, et al. (2024) **Web-based multi-omics integration using the Analyst software suite** *Nat Protoc* **19**:1467–97 <https://doi.org/10.1038/s41596-023-00950-4> | PubMed | Google Scholar
91. Ahmed Z, Wan S, Zhang F, Zhong W (2024) **Artificial intelligence for omics data analysis** *BMC Methods* **1**:4 <https://doi.org/10.1186/s44330-024-00004-5> | Google Scholar
92. Mann M, Kumar C, Zeng WF, Strauss MT (2021) **Artificial intelligence for proteomics and biomarker discovery** *Cell Syst* **12**:759–70 <https://doi.org/10.1016/j.cels.2021.06.006> | PubMed | Google Scholar
93. Toussaint PA, Leiser F, Thiebes S, Schlesner M, Brors B, Sunyaev A (2023) **Explainable artificial intelligence for omics data: a systematic mapping study** *Brief Bioinform* **25** <https://doi.org/10.1093/bib/bbad453> | PubMed | Google Scholar
94. Sun L, Guo W, Guo L, Chen X, Zhou H, Yan S, et al. (2024) **Molecular landscape and multi-omic measurements of heterogeneity in fetal adenocarcinoma of the lung** *NPJ Precis Oncol* **8**:99 <https://doi.org/10.1038/s41698-024-00569-y> | PubMed | Google Scholar
95. Halama A, Zaghlool S, Thareja G, Kader S, Al Muftah W, Mook-Kanamori M, et al. (2024) **A roadmap to the molecular human linking multiomics with population traits and diabetes subtypes** *Nature communications* **15**:7111 <https://doi.org/10.1038/s41467-024-51134-x> | PubMed | Google Scholar
96. Garg M, Karpinski M, Matelska D, Middleton L, Burren OS, Hu F, et al. (2024) **Disease prediction with multi-omics and biomarkers empowers case-control genetic discoveries in the UK Biobank** *Nat Genet* **56**:1821–31 <https://doi.org/10.1038/s41588-024-01898-1> | PubMed | Google Scholar
97. Reid M, Agbassi YJP, Arinaminpathy N, Bercasio A, Bhargava A, Bhargava M, et al. (2023) **Scientific advances and the end of tuberculosis: a report from the Lancet Commission on Tuberculosis** *Lancet* **402**:1473–98 [https://doi.org/10.1016/S0140-6736\(23\)01379-X](https://doi.org/10.1016/S0140-6736(23)01379-X) | PubMed | Google Scholar

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Reviewer #1 (Public review):

Summary:

This is an interesting and useful review highlighting the complex pathways through which pulmonary colonisation or infection with *Mycobacterium tuberculosis* (Mtb) may progress to

develop symptomatic disease and transmit the pathogen. I found the section on immune correlates associated with individuals who have clearly been exposed to and reacted to Mtb but did not develop latent infections particularly valuable. However, several aspects would benefit from clarification.

Strengths:

The main strengths lie in the arguments presented for a multiplicity of immune pathways to TB disease.

Weaknesses:

The main weaknesses lie in clarity, particularly in the precise meanings of the three figures.

I accept that there is a 'goldilocks zone' that underpins the majority of TB cases we see and predominantly reflects different patterns of immune response, but the analogies used need to be more clearly thought through.

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Reviewer #2 (Public review):

Summary:

This is a thought-provoking perspective by Reichmann et al, outlining supportive evidence that *Mycobacterium tuberculosis* co-evolved with its host *Homo Sapiens* to both increase susceptibility to infection and reduce rates of fatal disease through decreased virulence. TB is an ancient disease where two modes of virulence are likely to have evolved through different stages of human evolution: one before the Neolithic Demographic Transition, where humans lived in sparse hunter-gatherer communities, which likely selected for prolonged Mtb infection with reduced virulence to allow for transmission across sparse populations. Conversely, following the agricultural and industrial revolutions, Mtb virulence is likely to have evolved to attack a higher number of susceptible individuals. These different disease modalities highlight the central idea that there are different immunological routes to TB disease, which converge on a disease phenotype characterized by high bacterial load and destruction of the extracellular matrix. The writing is very clear and provides a lot of supportive evidence from population studies and the recent clinical trials of novel TB vaccines, like M72 and H56. However, there are areas to support the thesis that have been described only in broad strokes, including the impact of host and Mtb genetic heterogeneity on this selection, and the alternative model that there are likely different TB diseases (as opposed to different routes to the same disease), as described by several groups advancing the concept of heterogeneous TB endotypes. I expand on specific points below.

Strengths:

(1) The idea that Mtb evolved to both increase transmission (and possible commensalism with humans) with low rates of reactivation is intriguing. The heterogeneous TB phenotypes in the collaborative cross model (PMID: 35112666) support this idea, where some genetic backgrounds can tolerate a high bacterial load with minimal pathology, while others show signs of pathogenesis with low bacterial loads. This supports the idea that the underlying host state, driven by a number of factors like genetics and nutrition, is likely to explain whether someone will co-exist with Mtb without pathology, or progress to disease. I particularly enjoyed the discussion of the protective advantages provided by Mtb infection, which may have rewired the human immune system to provide protection against heterologous pathogens- this is supported by recent studies showing that Mtb infection provides moderate protection against SARS-CoV-2 (PMID: 35325013, and 37720210), and may have applied to

other viruses that are likely to have played a more significant role in the past in the natural selection of *Homo Sapiens*.

(2) Modeling from Marcel Behr and colleagues (PMID: 31649096) indeed suggests that there are at least TB clinical phenotypes that likely mirror the two distinct phases of *Mtb* co-evolution with humans. Most of the TB disease progression occurs rapidly (within 1-2 years of exposure), and the rest are slow cases of reactivation over time. I enjoyed the discussion of the difference between the types of immune hits needed to progress to disease in the two scenarios, where you may need severe immune hits for rapid progression, a phenotype that likely evolved after the Neolithic transition to larger human populations. On the other hand, a series of milder immune events leading to reactivation after a long period of asymptomatic infection likely mirrors slow progression in the hunter-gatherer communities, to allow for prolonged transmission in scarce populations. Perhaps a clearer analysis of these models would be helpful for the reader.

Weaknesses:

(1) The discussion of genetic heterogeneity is limited and only discusses evidence from MSMD studies. Genetics is an important angle to consider in the co-evolution of *Mtb* and humans. There is a large body of literature on both host and *Mtb* genetic associations with TB disease. The very fact that host variants in one population do not necessarily cross-validate across populations is evidence in support of population-specific adaptations. Specific *Mtb* lineages are likely to have co-evolved with distinct human populations. A key reference is missing (PMID: 23995134), which shows that different lineages co-evolved with human migrations. Also, meta-analyses of human GWAS studies to define variants associated with TB are very relevant to the topic of co-evolution (e.g., PMID: 38224499). eQTL studies can also highlight genetic variants associated with regulating key immune genes involved in the response to TB. The authors do mention that *Mtb* itself is relatively clonal with ~2K SNPs marking *Mtb* variation, much of which has likely evolved under the selection pressure of modern antibiotics. However, some of this limited universe of variants can still explain co-adaptations between distinct *Mtb* lineages and different human populations, as shown recently in the co-evolution of lineage 2 with a variant common in Peruvians (PMID: 39613754).

(2) Although the examples of anti-TNF and anti-PD1 treatments are relevant as drivers of TB in limited clinical contexts, the bigger picture is that they highlight major distinct disease endotypes. These restricted examples show that TB can be driven by immune deficiency (as in the case of anti-TNF, HIV, and malnutrition) or hyperactivation (as in the case of anti-PD1 treatment), but there are still certainly many other routes leading to immune suppression or hyperactivation. Considering the idea of hyper-activation as a TB driver, the apparent higher rate of recurrence in the H56 trial referenced in the review is likely due to immune hyperactivation, especially in the context of residual bacteria in the lung. These different TB manifestations (immune suppression vs immune hyperactivation) mirror TB endotypes described by DiNardo et al (PMID: 35169026) from analysis of extensive transcriptomic data, which indicate that it's not merely different routes leading to the same final endpoint of clinical disease, but rather multiple different disease endpoints. A similar scenario is shown in the transcriptomic signatures underlying disease progression in BCG-vaccinated infants, where two distinct clusters mirrored the hyperactivation and immune suppression phenotypes (PMID: 27183822). A discussion of how to think about translating the extensive information from system biology into treatment stratification approaches, or adjunct host-directed therapies, would be helpful.

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Reviewer #3 (Public review):**Summary:**

This perspective article by Reichmann et al. highlights the importance of moving beyond the search for a single, unified immune mechanism to explain host-Mtb interactions. Drawing from studies in immune profiling, host and bacterial genetics, the authors emphasize inconsistencies in the literature and argue for broader, more integrative models. Overall, the article is thought-provoking and well-articulated, raising a concept that is worth further exploration in the TB field.

Strengths:

Timely and relevant in the context of the rapidly expanding multi-omics datasets that provide unprecedented insights into host-Mtb interactions.

Weaknesses (Minor):

- (1) Clarity on the notion of a "unified mechanism". It remains unclear whether prior studies explicitly proposed a single unifying immunological model. While inconsistencies in findings exist, they do not necessarily demonstrate that earlier work was uniformly "single-minded". Moreover, heterogeneity in TB has been recognized previously (PMIDs: 19855401, 28736436), which the authors could acknowledge.
- (2) Evolutionary timeline and industrial-era framing. The evolutionary model is outdated. Ancient DNA studies place the Mtb's most recent common ancestor at ~6,000 years BP (PMIDs: 25141181; 25848958). The Industrial Revolution is cited as a driver of TB expansion, but this remains speculative without bacterial-genomics evidence and should be framed as a hypothesis. Additionally, the claim that Mtb genomes have been conserved only since the Industrial Revolution (lines 165-167) is inaccurate; conservation extends back to the MRCA (PMID: 31448322).
- (3) Trained immunity and TB infection. The treatment of trained immunity is incomplete. While BCG vaccination is known to induce trained immunity (ref 59), revaccination does not provide sustained protection (ref 8), and importantly, Mtb infection itself can also impart trained immunity (PMID: 33125891). Including these nuances would strengthen the discussion.

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Author response:

We thank the reviewers for their primarily positive comments and the critiques about where the manuscript could be improved. We agree with the vast majority of points raised. In our revised submission, we will:

- Clarify some of the wording such as "unified mechanism" so that our intended meaning is clear to all readers
- Completely change figure 2, as we accept the critique that an X-Y plot is not the logical way to present this concept
- Amend the legends of figures 1 and 3 so that the disease pathways we are attempting to illustrate are clear for all readers

- Expand on the genetic interactions between humans and TB and cite the manuscripts suggested
- Add further discussion on multiple disease endotypes, and the immunological events that may lead to these distinct end points, along with how this may inform treatment stratification approaches
- Extend the discussion about trained immunity
- Make specific changes to address each of the reviewers' points in the recommendations to authors
- In the minority of cases where we feel a change is not necessary, we will justify this in our response to reviews

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